```
# Curve fitting of a straight line:
import numpy as np
from scipy.optimize import curve_fit
import matplotlib.pyplot as plt

# Define the linear function y = a + bx
def linear_function(x,a,b):
  return(a + b * x)

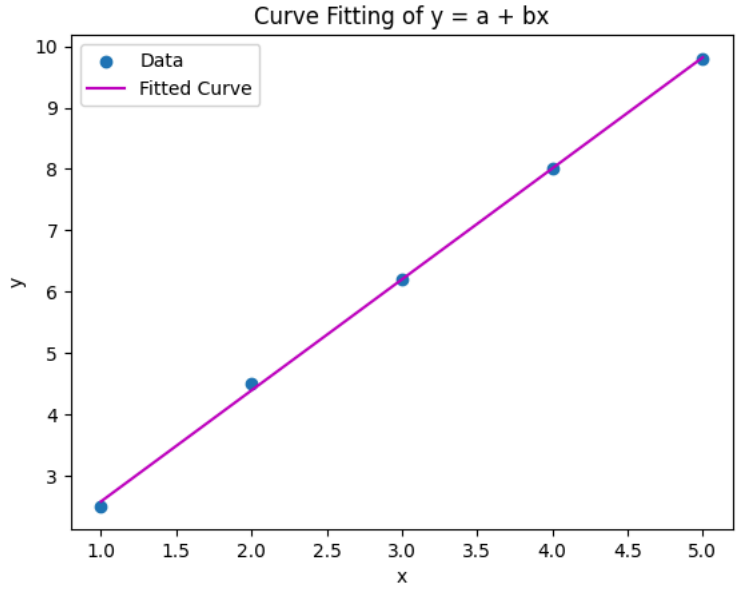
# Example data
x = np.array([1,2,3,4,5])
y = np.array([2.5,4.5,6.2,8.0,9.8])

# Perform curve fitting
Params, covariance = curve_fit(linear_function, x,y)
a_fit, b_fit = Params

# Print fitted parameters
print("Fitted a value:", a_fit)
print("Fitted b value:", b_fit)

# Create a scatter plot and plot the fitted curve
plt.scatter(x, y, label="Data")
plt.plot(x, linear_function(x, a_fit, b_fit), color="m", label="Fitted Curve")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Curve Fitting of y = a + bx")
plt.legend()
plt.show()
```

```
Fitted a value: 0.7699999999999995
Fitted b value: 1.81
```




```
# Curve fitting of a parabola:
import numpy as np
from scipy.optimize import curve_fit
import matplotlib.pyplot as plt

# Define the parabola function
def parabola (x,a,b,c):
  return(a + (b * x) +(c * x**2))

# Example data
x_data = np.array([1,2,3,4,5])
y_data = np.array([3.5,7.2,13.8,23.5,36.2])

# Fit the data to the parabola function
Params, covariance = curve_fit(parabola, x_data, y_data)

# Extract the fitted parameters
a_fit, b_fit, c_fit = Params

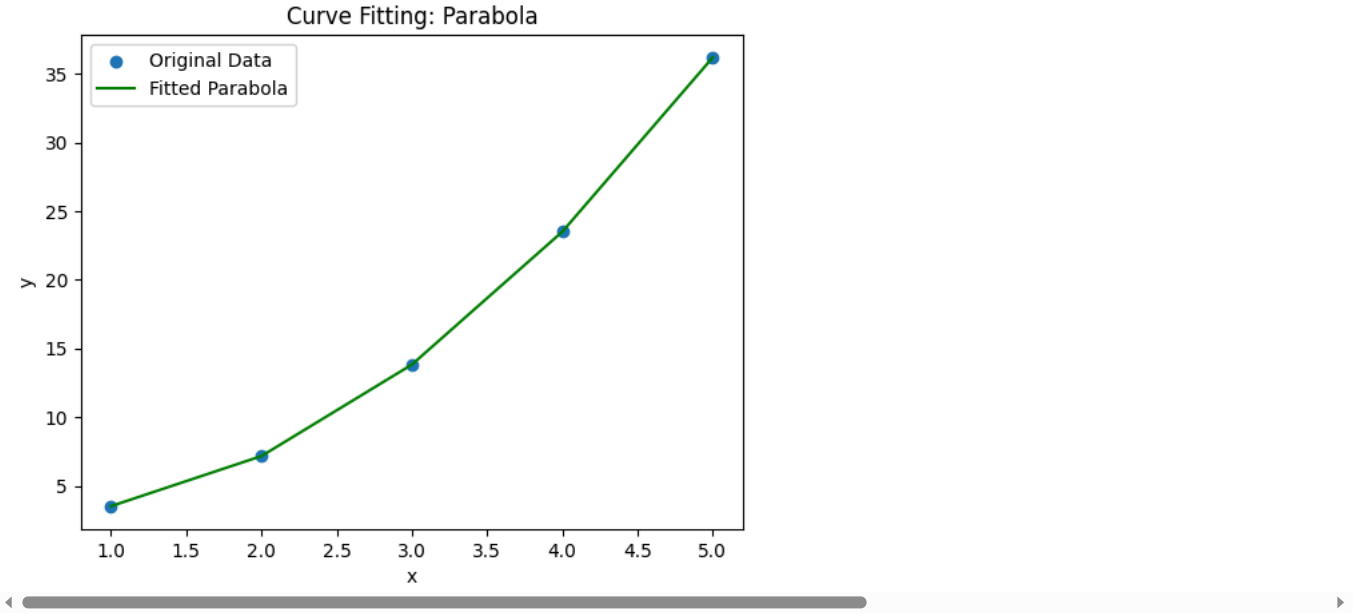
# Generate fitted y values
y_fit = parabola(x_data, a_fit, b_fit, c_fit)

# Print fitted parameters
print("Fitted a value:", a fit)
```

```
print("Fitted b value:", b_fit)
print("Fitted c value:", c_fit)

# Plot the original data and the fitted parabola
plt.scatter(x_data, y_data, label="Original Data")
plt.plot(x_data, y_fit, color="g", label="Fitted Parabola")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Curve Fitting: Parabola")
plt.legend()
plt.show()
```

```
Fitted a value: 2.8800000042752245
Fitted b value: -0.8728571457932627
Fitted c value: 1.5071428575807633
```




```
# Curve fitting of an exponential growth curve:
import numpy as np
from scipy.optimize import curve_fit
import matplotlib.pyplot as plt

# Exponential growth function
def exponential_growth(x,a,b):
  return(a * (x**b))

# Example data
x = np.array([1,2,3,4,5])
y = np.array([2,8,18,32,50])

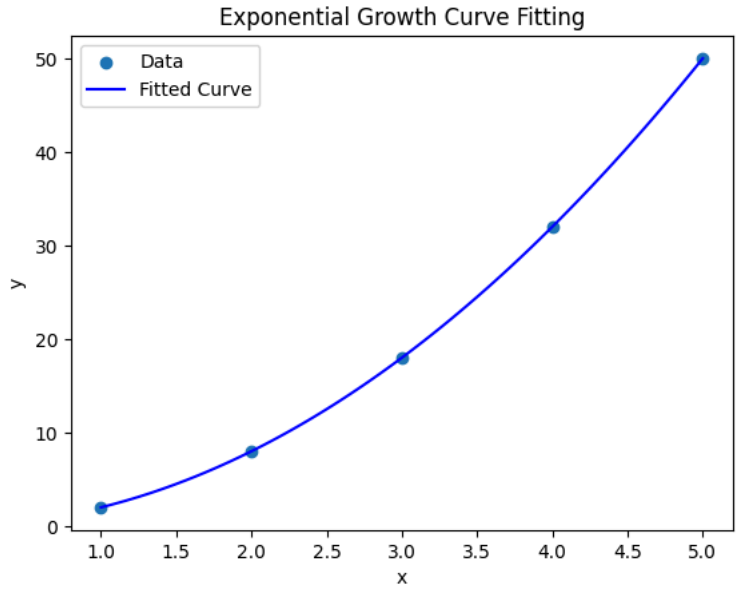
# Perform curve fitting
Params, covariance = curve_fit(exponential_growth,x, y)

# Extract fitted parameters
a_fit, b_fit = Params

# Generate fitted data points
x_fit = np.linspace(min(x),max(x),100)
y_fit = exponential_growth(x_fit, a_fit, b_fit)

# Create a scatter plot and plot the fitted curve
plt.scatter(x, y, label="Data")
plt.plot(x_fit, y_fit, color="b", label="Fitted Curve")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Exponential Growth Curve Fitting")
plt.legend()
plt.show()

print("Fitted parameters:")
print("Fitted a value:", a_fit)
print("Fitted b value:", b_fit)
print("Fitted c value:", c_fit)
```




```
Fitted parameters:
Fitted a value: 2.0
Fitted b value: 2.0
Fitted c value: 1.5071428575807633
```